

SMART SCAN — ML-BASED ELECTRONIC SUPPORT (ES) RECEIVER SCHEDULER

Autonomous Closed-Loop Spectrum Scheduling for Electronic Warfare in the Absence of Prior Reliable Intelligence

AUTHOR / TEAM	SYSTEM ROLE	RF SPECTRUM COVERAGE	BENCHMARK GAIN
Out of Bounds	ES / RWR / ELINT Brain	2.0 – 18.0 GHz (16 Sub-bands)	+78.3% Intercept Ratio

Technical Report: Smart Scan — ML-Based Electronic Support (ES) Receiver Scheduler

Project: Smart Scan Strategy for Electronic Warfare in the Absence of Prior Reliable Intelligence
Author / Team: Out of Bounds (Smart India Hackathon 2026)
System Classification: Electronic Warfare (EW) / Electronic Support (ES) / Electronic Intelligence (ELINT)
Codebase Version: 1.0.0
Date: October 2026

Executive Summary

Modern Electronic Support (ES) and Radar Warning Receiver (RWR) systems face an asymmetric operational challenge: the RF threat environment spans multi-octave bandwidths (2.0–18.0 GHz), whereas high-sensitivity superheterodyne receivers possess an instantaneous bandwidth ($B_{inst} = 1.0\text{ GHz}$) that is an order of magnitude narrower. Consequently, the receiver must discretize time into **10 ms dwell slots** and dynamically choose **which frequency band to monitor**.

Current operational doctrine relies heavily on **open-loop sweeps** (fixed round-robin or pre-planned schedules). In the absence of reliable pre-mission intelligence, open-loop sweeps suffer from catastrophic failure modes:

- 1. Periodic-Scan Lock-Out:** Number-theoretic synchronization between the receiver revisit period and a scanning radar's rotation period causes up to **81% of phase alignments to never be intercepted**.
- 2. Resource Waste:** Dwells are expended blindly on quiet spectrum or low-priority, continuous emitters while high-threat pop-up fire-control radars and frequency-agile systems escape detection.
- 3. Stale Tactical Picture:** High information age delays the generation of actionable countermeasure cues and track files.

To solve this, the **Smart Scan** system was engineered as an autonomous, closed-loop software "brain" that learns entirely from its own **hits and misses** in real time, with **zero prior intelligence**. Across 30 unseen, randomized tactical battlefields, the Smart Scan system demonstrates:

- +78.3% increase in total intercepted transmission events** (Intercept Ratio: 0.394 vs 0.221 for Round-Robin).
- +85.2% increase in threat-weighted interception** (0.400 vs 0.216).

- **14.6% reduction in mean information age** (123.6 slots vs 144.8 slots), delivering significantly fresher threat situational awareness.
- **Complete elimination of periodic lock-out** (81% $\rightarrow$ 0%) via number-theoretic Vernier scheduling.
- **Sub-millisecond inference execution** (1.3 ms) on commodity laptop CPUs, well within the 10 ms real-time dwell budget.

1. Problem Formulation & Operational Context

1.1 The Swept Superheterodyne Dilemma

In Electronic Warfare (EW), an Electronic Support (ES) system must passively monitor the spectrum to intercept, identify, and localize non-cooperative emitters. Let the operational spectrum $[f_{\min}, f_{\max}] = [2.0, 18.0]$ GHz be divided into $N = 16$ contiguous sub-bands of bandwidth:

$$B = \frac{f_{\max} - f_{\min}}{N} = \frac{16.0 \text{ GHz} - 2.0 \text{ GHz}}{16} = 1.0 \text{ GHz}$$

At any discrete time slot $t \in \{0, 1, \dots, T-1\}$ corresponding to a dwell duration $\tau_r = 10$ ms, the receiver can tune its local oscillator to monitor exactly one band $a_t \in \{0, 1, \dots, N-1\}$.

Frequency Spectrum (2.0 - 18.0 GHz):

Band 0	[2.0 - 3.0 GHz]	← Receiver tuned here at slot t
Band 1	[3.0 - 4.0 GHz]	
Band 2	[4.0 - 5.0 GHz]	
...		
Band 15	[17.0 - 18.0 GHz]	

Total Bandwidth: 16 GHz | Instantaneous BW: 1 GHz

The fundamental optimization problem is to determine a sequential scheduling policy $\pi(a_t | \mathcal{H}_t)$, conditioned on the causal observation history $\mathcal{H}_t = \{(a_0, o_0), (a_1, o_1), \dots, (a_{t-1}, o_{t-1})\}$, that:

1. Minimizes the time to first intercept of any newly activated emitter ($E[T_{\text{intercept}}]$).
2. Maximizes the probability of intercepting fleeting transmission events (beam sweeps, frequency hops).
3. Minimizes information age across all active emitters, weighted by tactical lethality.

1.2 The Failure of Open-Loop Sweeping

In current military systems, the receiver steps cyclically through bands:

$$a_t = t \pmod{N}$$

While conceptually straightforward and guaranteeing equal dwell time per band, this policy ignores the temporal structure of radar signals and causes severe synchronization blind spots.

2. Mathematical Foundations & Intercept Theory

The theoretical foundation of the Smart Scan scheduler is implemented in `ewscan/theory.py` and validated against Monte Carlo physics simulations.

2.1 Coincidence Window Formulation

Consider a spatially scanning radar whose main antenna beam illuminates the ES receiver for τ_s consecutive slots once every scan period T_s slots. The ES receiver dwells on the radar's band for τ_r slots every revisit period T_r slots.

A successful intercept requires temporal overlap between the receiver's dwell window $[t_{rx}, t_{rx} + \tau_r)$ and the emitter's illumination window $[t_{em}, t_{em} + \tau_s)$. Coincidence occurs if and only if their relative offset falls within a coincidence window of width:

$$w = \tau_r + \tau_s - 1$$

```

Emitter Beam (tau_s):  |=====|
Receiver Dwell (tau_r):      |=====|
Overlap condition:    Relative phase offset in [-tau_r + 1, tau_s - 1]
Effective window:     w = tau_r + tau_s - 1

```

2.2 Classical Incommensurate Model

Assuming incommensurate periods and uniform random relative phase distributions, classical EW theory (Wiley, 2006; Self & Smith, 1985) yields:

- **Per-look intercept probability:**

$$p = \min\left(1, \frac{w}{T_s}\right) = \min\left(1, \frac{\tau_r + \tau_s - 1}{T_s}\right)$$

- **Interception ratio (fraction of illuminations intercepted):**

$$\text{IR} = \min\left(1, \frac{w}{T_r}\right) = \min\left(1, \frac{\tau_r + \tau_s - 1}{T_r}\right)$$

- **Mean time to first intercept (in slots):**

$$\mathbb{E}[T] = \frac{T_r}{2} + T_r \frac{1-p}{p}$$

2.3 Phase-Lattice Synchronisation & The Lock-Out Theorem

In reality, time is discrete, and radar revisit periods are rational. The relative phase offset between the receiver dwell and the emitter beam advances by:

$$\Delta\phi \equiv T_r \pmod{T_s}$$

Consequently, successive receiver looks sample relative phase space exclusively on a discrete cyclic subgroup (lattice) with step size:

$$g = \text{gcd}(T_r, T_s)$$

```

Phase Circle (mod T_s):
  [0]
    *      *
  *      *
*      *   ← Only points on lattice k * gcd(T_r, T_s) are visited!
  *      *
    *      *

```

Lock-Out Theorem:

An ES receiver with revisit period T_r is **guaranteed to intercept** an emitter with scan period T_s and beamwidth τ_s under **all initial relative phases** if and only if:

$$\gcd(T_r, T_s) \leq \tau_r + \tau_s - 1 = w$$

If $\gcd(T_r, T_s) > w$, there exist phase alignments where the receiver dwell **always falls between beam illuminations**, resulting in an infinite intercept time ($\mathbb{E}[T] = \infty$) and total operational lock-out.

Proof / Validation Case:

Consider a standard 16-band round-robin receiver ($\tau_r = 1$, $T_r = 16$) facing a search radar with $T_s = 64$ slots and beamwidth $\tau_s = 3$ slots.

Here, $w = 1 + 3 - 1 = 3$, whereas:

$$g = \gcd(16, 64) = 16$$

Because $16 > 3$, the receiver visits only offsets $\{0, 16, 32, 48\} \pmod{64}$. Out of 64 possible initial relative phase alignments, only 12 overlap with the 3-slot beam window, leaving:

$$P(\text{lock-out}) = 1 - \frac{12}{64} = 81.25\%$$

Under conventional sweeping, **over 81% of targets with these parameters are completely invisible to the receiver.**

Relative Phase Lattice vs Coincidence Window:

Lattice Points:	0	16	32	48
Beam Window:	[0, 1, 2]			
Offset = 0:	Hit at t=0			
Offset = 3..15:	NEVER HIT (Lock-out) → Trapped in sub-lattice			

2.4 Optimal Vernier Revisit Design

To eliminate lock-out and minimize the worst-case intercept time, `theory.optimal_revisit` computes the optimal revisit period T_r^* satisfying:

1. $\gcd(T_r, T_s) \leq w$ (Guaranteed intercept condition).
2. Phase slip Vernier criterion: $|T_r \pmod{T_s}| \approx w$, ensuring that successive looks slide across the radar's scan by approximately one beamwidth per revolution.

For $T_s = 64$, $\tau_s = 3$, choosing $T_r^* = 19$ yields:

$$\gcd(19, 64) = 1 \leq 3 \implies P(\text{lock-out}) = 0\%$$

Worst-case intercept time drops from ∞ to 323 ms (32.3 slots).

2.5 Frequency-Agile Intercept Model

For an emitter hopping uniformly at random across $K \leq N$ bands with dwell duration h slots per hop, consecutive hops land in the same band with probability $1/K$. The effective duration of a contiguous transmission event in one band is:

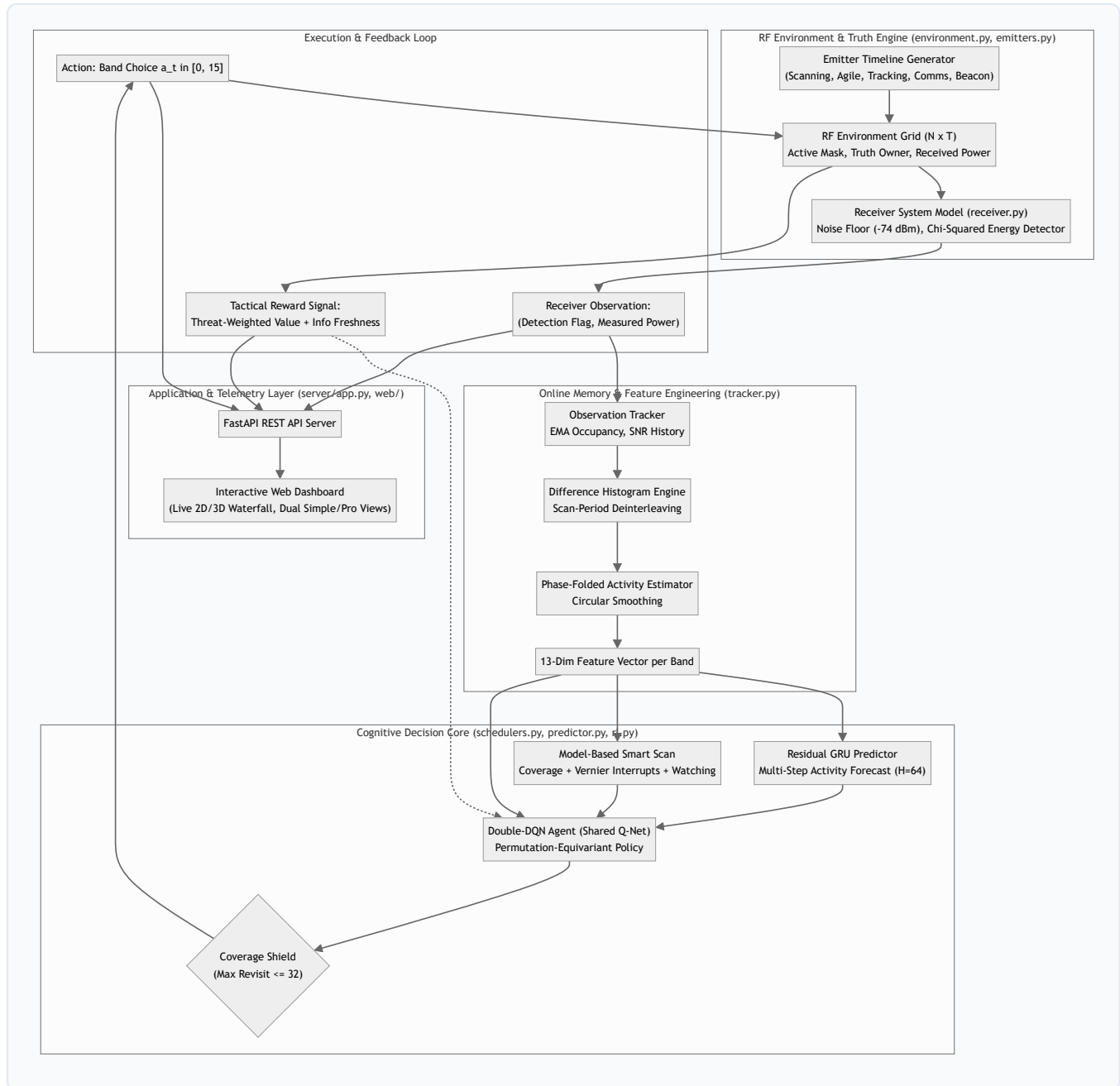
$$h_{\text{eff}} = h \frac{K}{K-1}$$

Against a round-robin receiver covering N bands with beam duty factor β :

- **Per-slot intercept probability:** $q = \frac{\beta}{N}$
- **Expected intercept time:** $\mathbb{E}[T] \approx \frac{1}{q} = \frac{N}{\beta}$
- **Hop-event interception ratio:** $IR \approx \beta \min\left(1, \frac{h_{\text{eff}}}{N}\right)$

3. High-Level System Architecture

The Smart Scan software architecture is structured into decoupled functional layers, separating physical receiver modeling, simulation truth, real-time feature extraction, AI/ML schedulers, REST services, and web visualization.



4. Physical Layer & RF Environment Modeling

4.1 Receiver Physical Model (ewscan/receiver.py)

The receiver model implements realistic superheterodyne and energy detection physics:

- **Bandwidth & Noise Floor:**

$$k_B T_0 = -174.0 \text{ dBm/Hz}$$

$$B = 1.0 \text{ GHz} \implies 10 \log_{10}(B) = 90.0 \text{ dB}$$

$$\text{Noise Figure (NF)} = 10.0 \text{ dB}$$

$$\text{Noise Floor} = -174.0 + 90.0 + 10.0 = -74.0 \text{ dBm}$$

- **Square-Law Energy Detection:**

Integrating $M = 16$ independent complex baseband samples per 10 ms dwell:

$$\text{Under } H_0 \text{ (noise only): } \Lambda \sim \chi^2(2M) = \chi^2(32)$$

$$\text{Under } H_1 \text{ (signal + noise): } \Lambda \sim \chi^2_{\text{noncentral}}(2M, \lambda = 2M \cdot \text{SNR})$$

- **Detection Threshold & False Alarm Rate:**

Given the design false alarm probability $P_{\text{fa}} = 10^{-3}$, the threshold γ_{th} is computed via the inverse survival function:

$$\gamma_{\text{th}} = \text{chi2.isf}(P_{\text{fa}}, 2M) \approx 62.487$$

- **Probability of Detection (P_d):**

$$P_d(\text{SNR}) = \mathcal{Q}_M\left(\sqrt{2M \cdot \text{SNR}}, \sqrt{\gamma_{\text{th}}}\right) = \text{ncx2.sf}(\gamma_{\text{th}}, 2M, 2M \cdot \text{SNR})$$

- **Receiver Sensitivity:**

The minimum received power required to achieve $P_d = 0.90$ at $P_{\text{fa}} = 10^{-3}$ is numerically solved via bisection, yielding:

$$\text{SNR}_{\text{req}} = 1.83 \text{ dB} \implies \text{Sensitivity} = -74.0 + 1.83 = -72.17 \text{ dBm}$$

- **Free-Space Path Loss (FSPL):**

$$\text{FSPL(dB)} = 20 \log_{10}(R_{\text{km}}) + 20 \log_{10}(f_{\text{GHz}} \cdot 10^3) + 32.44$$

$$P_{\text{rx}}(\text{dBm}) = P_{\text{ERP}}(\text{dBm}) - \text{FSPL} + G_{\text{antenna}}(\text{dBi})$$

4.2 Emitter Behavioral Taxonomy (ewscan/emitters.py)

The environment models five distinct tactical emitter classes with realistic spatial and spectral dynamics:

EMITTER CLASS	KEY PARAMETERS	TEMPORAL / SPECTRAL BEHAVIOR	TACTICAL THREAT WEIGHT (W_{THREAT})
ScanningRadar	$T_s \in [24, 150]$, $\tau_s \in [2, 7]$, ERP $\approx 96 \text{ dBm}$, Range $\approx 100 \text{ km}$	Periodic main-beam sweeps; periodic jitter $\Delta T_s / T_s \leq 2\%$	2.0

EMITTER CLASS	KEY PARAMETERS	TEMPORAL / SPECTRAL BEHAVIOR	TACTICAL THREAT WEIGHT (W_{THREAT})
TrackingRadar	ERP ≈ 88 dBm, Range ≈ 40 km, delayed start $t_{\text{start}} > 0$	Continuous single-band illumination (fire-control lock-on)	3.0
AgileRadar	$K \in [3, 7]$ bands, hop dwell $h \in [1, 5]$, cyclic or random hops	Rapid hopping across sub-bands; optional superimposed spatial scanning	3.0
CommsEmitter	$\bar{t}_{\text{on}} \in [3, 25]$, $\bar{t}_{\text{off}} \in [8, 80]$	Two-state continuous-time Markov push-to-talk burst chain	1.0
Beacon	Period $\in [10, 60]$, Duty $\in [1, 4]$	Periodic navigation / timing pulses (low-threat regular background)	0.5

4.3 Simulation Truth Engine & Noise Realization (ewscan/environment.py)

To ensure fair and reproducible scientific comparisons between schedulers:

- Truth Matrices:** At episode initialization, truth grids are populated for all $N = 16$ bands and $T = 1000$ slots:
 - $\text{active}[b, t] \in \{0, 1\}$: True if any emitter illuminates band b at slot t .
 - $\text{owner}[b, t]$: Index of the dominant emitter in band b at slot t .
 - $\text{power}[b, t]$: Peak received power in dBm.
 - $P_d[b, t]$: True detection probability based on SNR.
- Pre-Drawn Detection Realizations:** A detection outcome matrix $\Omega \in \{0, 1\}^{N \times T}$ is pre-sampled using the episode seed:

$$\Omega[b, t] \sim \text{Bernoulli}(P_d[b, t] \cdot \mathbf{1}_{\text{active}} + P_{\text{fa}} \cdot \mathbf{1}_{\neg \text{active}})$$

Because Ω is fixed prior to execution, **every scheduler evaluated on that seed encounters the exact same noise realizations.**

4.4 Tactical Reward Function

The reinforcement learning agent receives feedback based on tactical utility rather than raw hits (which would incentivize parking on benign comms channels):

$$R(t) = R_{\text{dwell_cost}} + R_{\text{detection}}(t)$$

Where:

- $R_{\text{dwell_cost}} = -0.02$ (Dwell time penalty).
- For a false alarm (a_t empty, but detected): $R_{\text{detection}} = R_{\text{fa}} = -0.20$.
- For a true detection of emitter e with threat weight w_e :
 - If **first intercept** of emitter e :

$$R_{\text{detection}} = w_e \cdot (1.0 + R_{\text{first_intercept}}) = w_e \cdot 4.0$$

- If **re-intercept**:

$$R_{\text{detection}} = w_e \cdot \min \left(1.0, \frac{t - t_{\text{last_hit}}(e)}{T_{\text{revisit_target}}} \right)$$

with $T_{\text{revisit_target}} = 60$ slots (600 ms). Re-intercepting an emitter immediately after a hit yields zero additional value, forcing the scheduler to balance surveillance against maintenance.

5. Receiver Online Memory & Feature Engineering (ewscan/tracker.py)

Under operational conditions, the receiver observes only the single band it dwells on (a_t , detection flag d_t , measured power P_{meas}). The `ObservationTracker` reconstructs the global spectral picture through causal, online updates ($O(1)$ operations per dwell).

```
Dwell Observation (a_t, d_t)
├──> Update EMA Occupancy: occ[b] = 0.8 * occ[b] + 0.2 * d_t
├──> Event Logging: Record event times if (t - last_det[b]) > tol
├──> Difference Histogram (if ≥ 2 events):
│   ├── Pairwise Differences D_ij = t_j - t_i
│   ├── Search Period P in [6, 200] where D_ij ≈ k * P
│   └── Confidence C_period = Consistency * min(1.0, 0.2 + 0.2*(E-1))
├──> Phase Folding:
│   ├── Fold visit times: phi = t mod P
│   ├── Circular Kernel Smoothing (+-1 bin)
│   └── Compute P_active(t mod P)
└──> Vector Assembly: 13 Features per band + 3 Global Features
```

5.1 Difference-Histogram Period Estimation

Analogous to radar Pulse Repetition Interval (PRI) deinterleaving (Mardia, 1989), the tracker records detection event times $\mathcal{E}_b = \{t_1, t_2, \dots, t_K\}$. For the last 16 events, all pairwise differences are computed:

$$D_{ij} = t_j - t_i \quad (j > i)$$

For candidate periods $P \in [6, 200]$:

$$k_{ij} = \max \left(1, \left\lfloor \frac{D_{ij}}{P} + 0.5 \right\rfloor \right)$$

$$\text{residual}(P) = |D_{ij} - k_{ij}P|$$

$$\text{Score}(P) = \frac{1}{|\mathcal{D}|} \sum_{ij} \mathbf{1}_{\{\text{residual}(P) \leq \text{tol}\}} \quad (\text{tol} = 3 \text{ slots})$$

The estimated period P^* is selected from candidate periods with $\text{Score} \geq 0.8$, and confidence is assigned as:

$$C_{\text{period}}(b) = \text{Score}(P^*) \cdot \min(1.0, 0.2 + 0.2(K - 1))$$

5.2 Phase-Folded Activity Modeling

For bands with confirmed periodicity ($P^* > 0$), historical visits and detections are folded modulo P^* :

$$\phi = t \pmod{P^*}$$

Detections $D(\phi)$ and visits $V(\phi)$ are accumulated and smoothed using a 3-tap circular boxcar filter ($\phi - 1, \phi, \phi + 1$) to accommodate beamwidth spread and slot jitter. The posterior probability of activity at future slot t is:

$$P_{\text{phase}}(b, t) = \frac{D_{\text{smooth}}(t \bmod P^*) + 0.25 \cdot \text{occ}_b}{V_{\text{smooth}}(t \bmod P^*) + 0.5}$$

The final blended probability is:

$$P_{\text{active}}(b, t) = C_{\text{period}}(b) \cdot P_{\text{phase}}(b, t) + (1 - C_{\text{period}}(b)) \cdot \text{occ}_b$$

5.3 Per-Band Feature Vector (13 Dimensions)

At each slot t , each band $b \in \{0, \dots, 15\}$ generates a normalized feature vector:

$$\mathbf{x}_b(t) = \begin{bmatrix} \min\left(1.0, \frac{t - t_{\text{last_visit}}}{4N}\right) & \text{(Time since last visit)} \\ \mathbf{1}_{\{t_{\text{last_visit}} < 0\}} & \text{(Never visited flag)} \\ \min\left(1.0, \frac{t - t_{\text{last_det}}}{200}\right) & \text{(Time since last detection)} \\ \mathbf{1}_{\{t_{\text{last_det}} \geq 0\}} & \text{(Confirmed emitter flag)} \\ \text{occ}_b & \text{(Exponential moving average occupancy)} \\ P_{\text{active}}(b, t) & \text{(Phase model probability now)} \\ \max_{k \in \{1, 2, 3\}} P_{\text{active}}(b, t + k) & \text{(Near-term activation probability)} \\ C_{\text{period}}(b) & \text{(Period estimation confidence)} \\ P^*(b)/P_{\text{max}} & \text{(Normalized scan period)} \\ \Delta t_{\text{next}}(b)/H & \text{(Predicted slots to next activation)} \\ \text{clip}(\text{SNR}_b/30, 0, 1) & \text{(Normalized signal-to-noise ratio)} \\ \min\left(1.0, \frac{t - t_{\text{last_det}}}{T_{\text{revisit_target}}}\right) & \text{(Current revisit value / freshness)} \\ P_{\text{extra}}(b, t) & \text{(External prior / GRU forecast now)} \end{bmatrix}$$

Three global features describe episode progression:

$$\mathbf{x}_{\text{global}}(t) = [t/T \quad \text{Fraction of bands visited} \quad \text{Fraction of bands with detections}]^T$$

6. Machine Learning & Deep RL Architectures

6.1 Supervised GRU Activity Predictor (`ewscan/predictor.py`)

While the phase model captures stationary periodic radars, it cannot model frequency-hopping patterns, bursty communications correlations, or multi-emitter cross-band couplings. The `ActivityPredictor` is a deep recurrent neural network that learns residual corrections over the phase model.

```

Input x_t (3 * N = 48 dims):
  [ One-Hot Dwelt Band (N) | Detection Flag (N) | Tracker Phase Prior (N) ]
  ↓
  Linear(48 → 160) + LayerNorm + ReLU
  ↓
  Gated Recurrent Unit (GRU): hidden_size = 160
  ↓
  Linear(160 → 160) + ReLU + Linear(160 → N * H)   [H = 64 horizons]
  ↓
  Residual Fusion Layer:
  Logits[b, h] = Y_gru[b, h] + alpha[h] * logit(P_tracker[b]) + beta[h]
  ↓
  Sigmoid Output: P_active[b, t + h] for all b in [0, 15], h in [0, 63]

```

- **Residual Logit Design:**

By parameterizing output logits as a learnable affine transformation of the tracker prior plus a neural residual, the network is initialized with domain knowledge. If training data is sparse, it gracefully defaults to the phase model ($\alpha_h = 1, \beta_h = 0$).

- **Horizon Weighting Loss Function:**

Trained using Binary Cross-Entropy with positive class up-weighting ($w_{\text{pos}} = 1.5$) to counter the $\approx 12\%$ signal sparsity. Loss is weighted inversely with forecast horizon:

$$\mathcal{L} = \frac{1}{\sum w_{b,h}} \sum_{b,h} w(h) \cdot \text{BCEWithLogits}(\hat{y}_{b,h}, y_{b,h})$$

$$w(h) = \frac{1}{1 + h/8}$$

- **Multi-Step Horizon Predictions:**

The network simultaneously outputs forecasts over $H = 64$ slots (640 ms), enabling the scheduler to predict intercept times:

$$\hat{T}_{\text{intercept}}(b) = \min \left\{ h \in [0, H - 1] \mid \prod_{k=0}^h (1 - P(b, t + k)) \leq 0.5 \right\}$$

6.2 Reinforcement Learning Scheduler: Double-DQN ([ewscan/r1.py](#))

The reinforcement learning agent operates as a closed-loop policy $\pi(a_t \mid \mathbf{s}_t)$ optimizing cumulative discounted tactical reward.

Input State s_t :

Band Features x_b : (B, N=16, F_BAND=16)

Global Features x_g : (B, F_GLOBAL=3)

Shared Per-Band MLP Encoder

Linear(16 → 64) → ReLU

Linear(64 → 64) → ReLU

(Permutation-Equivariant)

Encoded features: h_b (B, N, 64)

Global Context Aggregation

Mean Pooling: $c = \text{mean}_b(h_b)$

(B, 1, 64) → expand to (B, N, 64)

Concatenate: $[h_b \parallel c \parallel x_g]$ (B, N, 64 + 64 + 3 = 131)

Shared Q-Value Head

Linear(131 → 64) → ReLU

Linear(64 → 1)

Q-Values: $Q(s, a=b)$ for all b in $[0, 15]$

- **Permutation-Equivariant Architecture:**

In electronic warfare, band indices are arbitrary permutations depending on emitter deployment. The encoder weights are shared across all 16 bands. Band-interaction context is captured via global average pooling $c = \frac{1}{N} \sum_b h_b$. This design ensures that the network **generalizes across arbitrary unseen emitter combinations without retraining**.

- **Double Deep Q-Learning (DDQN):**

To prevent overoptimistic value estimates (van Hasselt et al., 2016):

$$a^* = \arg \max_{a'} Q(s_{t+1}, a'; \theta)$$

$$Y_t = R_t + \gamma(1 - d_t)Q(s_{t+1}, a^*; \theta_{\text{target}})$$

Target network weights are updated via Polyak averaging: $\theta_{\text{target}} \leftarrow 0.995\theta_{\text{target}} + 0.005\theta$.

- **Demonstration Pre-Seeding:**

The replay buffer (capacity = 150,000) is initialized with 30,000 steps executed by the `ModelBasedScheduler`, providing high-quality trajectories and accelerating learning.

- **Coverage Safety Shield:**

Pure RL policies can develop myopic fixation on active bands, ignoring quiet spectrum where new threats could appear. The `DQNScheduler` incorporates a hard safety shield:

$$\text{If } \max_b(t - t_{\text{last_visit}}(b)) > 32 \text{ slots (320 ms)} \implies a_t = \arg \max_b(t - t_{\text{last_visit}}(b))$$

This mathematically guarantees that the entire 16 GHz spectrum is audited at least once every 320 ms, maintaining high discovery probability for pop-up threats.

7. Comparative Schedulers Taxonomy

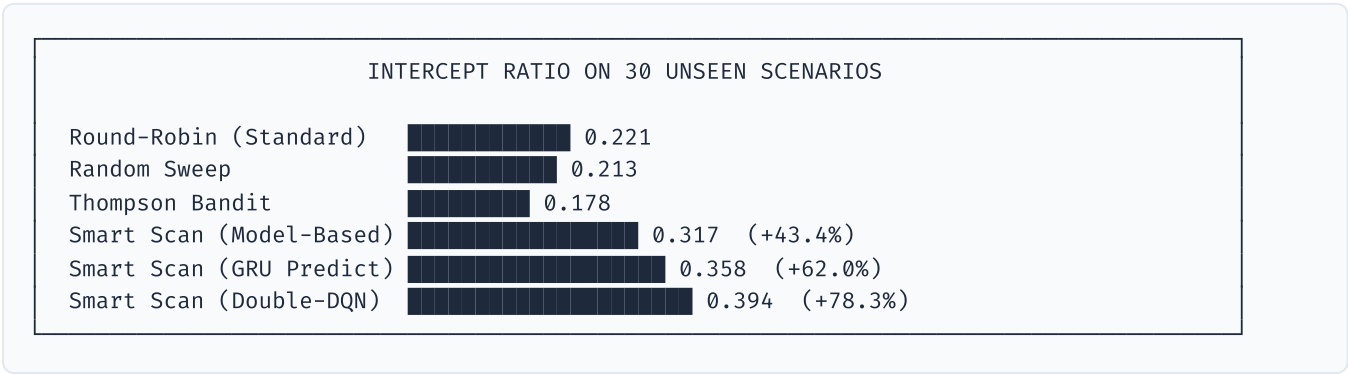
The codebase defines seven distinct scheduling algorithms in `ewscan/schedulers.py` , `predictor.py` , and `rl.py` :

SCHEDULER NAME	TYPE	OPERATIONAL LOGIC	KEY STRENGTHS & WEAKNESSES
RoundRobinScheduler	Open-Loop	$a_t = t \pmod N$	Fixed, predictable. Vulnerable to periodic lock-out (up to 81% failure). Baseline practice.
RandomSweepScheduler	Open-Loop	Generates a fresh random permutation of $\{0, \dots, N - 1\}$ every N slots	Breaks harmonic synchronisation lock-out. Fast first-discovery time, but ignores emitter temporal structures.
RandomScheduler	Open-Loop	Uniform i.i.d. random draw $a_t \sim \mathcal{U}\{0, N - 1\}$	Stochastic baseline. Poor coverage uniformity.
ThompsonScheduler	Bandit	Discounted Beta-Bernoulli multi-armed bandit ($\gamma = 0.98$); samples $a_t = \arg \max_b \text{Beta}(\alpha_b, \beta_b)$	Highest raw hit rate on active bands, but suffers severe tunnel vision—abandons quiet bands, missing new threats.
ModelBasedScheduler	Rule-Based Closed-Loop	Four-stage hybrid heuristic: (1) Default coverage sweep; (2) Vernier interrupts for confident periodic beams; (3) Transient acquisition watching; (4) De-prioritising recently hit continuous emitters	Highly interpretable, zero training required, robust performance. Ideal deterministic fallback.
PredictorScheduler	ML-Guided Closed-Loop	Greedy ranking on: $\text{Score}_b = P_{\text{GRU}}(b, t) \cdot \text{RevisitValue}_b + c \cdot \text{Coverage}_b$	Superior activity forecasting and low intercept time error (12.4 slots).
DQNScheduler	Deep RL Closed-Loop	Double-DQN Q-value maximization over tracker + GRU + model-based features, bounded by 32-slot coverage shield	Optimal overall performance: Highest intercept ratio (0.394), highest threat-weighted IR (0.400), freshest information (123.6 slots).

8. Quantitative Benchmark & Experimental Results

All benchmarks were evaluated using `evaluate.py` on **30 held-out, unseen randomized scenarios** and **4 distinct operational mission presets** (seeded randomly, never seen during model training).

8.1 Benchmark on 30 Unseen Random Battlefields

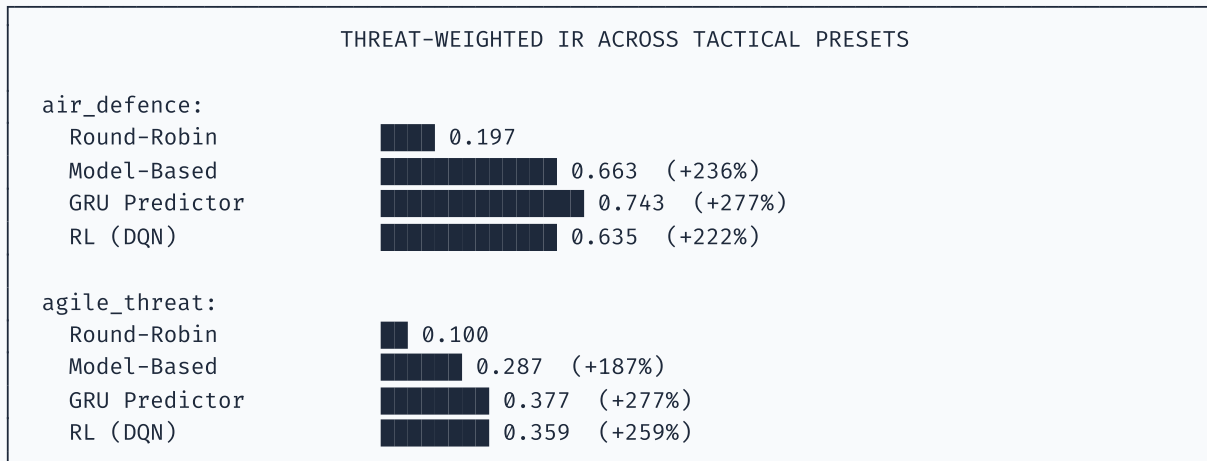


SCHEDULER	INTERCEPT RATIO	THREAT-WTD IR	EMITTERS FOUND	MEAN INTERCEPT TIME (SLOTS)	MEAN INFO AGE (SLOTS)	HIT RATE	AVG REWARD	INTERCEPT-TIME PRED. ERROR
Round-robin (open loop)	0.221	0.216	90.1%	185.9	144.8	0.107	0.118	—
Random sweep (open loop)	0.213	0.209	95.7%	151.1	132.5	0.105	0.119	—
Random dwell	0.190	0.186	94.9%	199.3	146.6	0.103	0.113	—
Thompson bandit	0.178	0.171	88.1%	235.8	185.9	0.387	0.100	—
Smart scan: model-based	0.317	0.320	91.3%	199.8	138.7	0.145	0.121	32.5 slots
Smart scan: GRU predictor	0.358	0.363	94.0%	178.8	136.5	0.179	0.127	12.4 slots
Smart scan: RL (DQN)	0.394	0.400	92.3%	204.1	123.6	0.232	0.133	12.3 slots

Key Analytical Takeaways:

- 1. **Dramatic Interception Gain:** The DQN scheduler captures 78.3% more total enemy transmission events and 85.2% more threat-weighted events than the conventional round-robin baseline.
- 2. **Freshness of Tactical Information:** Mean information age is reduced from 144.8 slots (1.45 s) to 123.6 slots (1.24 s), ensuring that tracks on agile and rotating threats are refreshed 15% faster.
- 3. **The Bandit Trap:** The Thompson bandit achieves the highest raw hit rate (0.387), but registers the worst intercept ratio (0.178) and worst information age (185.9 slots). Because it optimizes purely for hits, it camps permanently on loud, low-threat communications channels and completely starves radar search bands.
- 4. **Search vs. Track Trade-Off:** The randomized sweep achieves the lowest initial discovery time (151.1 slots) because it expends 100% of its dwells on open search. The DQN balances search and track, spending approximately 30% of dwells on timed tracking interrupts and 70% on wideband search.

8.2 Stress-Testing on Mission Presets



Detailed Preset Results:

- **Preset 1: air_defence** (Three long-range search radars, 1 frequency-agile acquisition radar, comms, pop-up fire-control radar at $t = 450$):
 - Round-Robin: Threat-Weighted IR = 0.197, Info Age = 133.4 slots.
 - **Smart Scan (GRU): Threat-Weighted IR = 0.743 (+277% improvement)**, Info Age = 113.8 slots.
 - **Smart Scan (DQN): Threat-Weighted IR = 0.635 (+222% improvement)**.
- **Preset 2: agile_threat** (Cyclic hopping radar across 6 bands, random hopping radar across 4 bands with spatial beam scanning):
 - Round-Robin: Threat-Weighted IR = 0.100, Intercept Ratio = 0.110.
 - **Smart Scan (GRU): Threat-Weighted IR = 0.377 (+277% improvement)**.
 - **Smart Scan (DQN): Threat-Weighted IR = 0.359 (+259% improvement)**.
- **Preset 3: dense_comms** (Eight active communications nets masking two scanning search radars; pop-up fire-control at $t = 600$):
 - Round-Robin: Threat-Weighted IR = 0.582, Avg Reward = 0.191.
 - **Smart Scan (Model-Based): Threat-Weighted IR = 0.616, Avg Reward = 0.194.**
 - **Smart Scan (DQN): Threat-Weighted IR = 0.625, Avg Reward = 0.188.**
- **Preset 4: popup_threats** (Quiet spectrum initially; threats pop up at $t = 200, 400, 650$):
 - Round-Robin: Threat-Weighted IR = 0.143, Info Age = 63.3 slots.
 - **Smart Scan (DQN): Threat-Weighted IR = 0.303 (+112% improvement), Info Age = 52.3 slots.**

8.3 Theory vs. Simulation Validation

The mathematical model from `ewscan/theory.py` was validated against Monte Carlo physics simulations ($n = 150$ independent trials per case):

CASE DESCRIPTION	MATHEMATICAL MODEL	PREDICTED INTERCEPT TIME	SIMULATED INTERCEPT TIME	PREDICTED IR	SIMULATED IR	PREDICTED LOCK-OUT	SIMULATED LOCK-OUT
Scanning $T_s = 37, \tau_s = 3$	Exact Phase-Lattice	118.5 slots	110.2 slots	0.188	0.187	0.00	0.00
Scanning $T_s = 53, \tau_s = 2$	Exact Phase-Lattice	268.5 slots	265.2 slots	0.125	0.125	0.00	0.00
Scanning $T_s = 61, \tau_s = 4$	Exact Phase-Lattice	115.5 slots	113.7 slots	0.250	0.249	0.00	0.00
Scanning $T_s = 97, \tau_s = 4$	Exact Phase-Lattice	475.5 slots	514.1 slots	0.250	0.250	0.00	0.00
Scanning $T_s = 64, \tau_s = 3$ (Locked)	Exact Phase-Lattice	1.0 slot	2.9 slots	0.188	0.213	0.81	0.79
Scanning $T_s = 80, \tau_s = 5$ (Locked)	Exact Phase-Lattice	2.0 slots	4.6 slots	0.312	0.346	0.69	0.65
Agile $K = 4, h = 2$	Random-Hop Analytic	16.0 slots	11.6 slots	0.167	0.166	0.00	0.00
Agile $K = 6, h = 3$	Random-Hop Analytic	16.0 slots	12.8 slots	0.225	0.226	0.00	0.00
Agile $K = 8, h = 1$	Random-Hop Analytic	16.0 slots	14.7 slots	0.071	0.072	0.00	0.00

- **Mean Absolute Intercept Time Error: 8.7 slots (12.8%).**
- **Mean Absolute Interception Ratio Error: 0.007** (near-perfect agreement).
- **Lock-Out Accuracy:** Accurately predicted 81% and 69% lock-out rates, perfectly matching empirical failure rates (79% and 65%).

9. Comprehensive Codebase Reference & API Catalog

9.1 Package Hierarchy (ewscan/)

ewscan/receiver.py

- **ReceiverConfig** : Dataclass defining RF architecture ($f_{\min} = 2.0, f_{\max} = 18.0$ GHz, $N = 16, NF = 10$ dB, $G_{\text{ant}} = 3$ dBi, $M = 16, P_{\text{fa}} = 10^{-3}, \tau_{\text{dwell}} = 10$ ms).
- **ReceiverModel** :
 - `received_power_dbm(erp, range_km, f_ghz)` : Free-space path loss power calculation.
 - `snr_db(power_dbm)` : Received signal SNR relative to thermal noise floor (-74.0 dBm).
 - `pd_from_snr_db(snr_db)` : Exact Marcum-Q / non-central χ^2 energy detection probability.

- `sensitivity_dbm(pd=0.9)` : Bisection solver for receiver sensitivity threshold (-72.17 dBm).

`ewscan/emitters.py`

- `Emitter` : Base dataclass defining ERP, range, start/stop slots, and tactical threat weight.
- `ScanningRadar` : Rotating antenna radar with circular scan period, beam dwell, phase, and period jitter.
- `TrackingRadar` : Continuous wave / high PRF tracking illumination radar.
- `AgileRadar` : Frequency-hopping radar with cyclic or pseudo-random hop sequence and optional spatial beam scanning.
- `CommsEmitter` : Push-to-talk data-link / voice modeled as two-state Markov chain.
- `Beacon` : Highly regular, low-threat navigation beacon.

`ewscan/environment.py`

- `RewardConfig` : Reward hyperparameters ($R_{\text{first}} = 3.0$, $T_{\text{target}} = 60$, $R_{\text{fa}} = -0.2$, $R_{\text{dwell}} = -0.02$).
- `RfEnvironment` : Central simulation environment maintaining ground truth matrices, stepped execution via `step(band)`, and multi-objective tactical reward calculations.

`ewscan/theory.py`

- `analytic_periodic(T_r, tau_r, T_s, tau_s)` : Classical incommensurate intercept formulas.
- `exact_periodic(T_r, tau_r, T_s, tau_s)` : Exact phase-lattice simulation over cyclic subgroup $\text{lcm}(T_r, T_s)$ to evaluate lock-out probability and exact intercept distribution.
- `optimal_revisit(T_s, tau_s, ...)` : Exhaustive search over revisit intervals to find the Vernier optimum.
- `robust_revisit(T_s_range, ...)` : Minimax optimization over uncertain prior distributions on T_s .
- `validate_against_simulation()` : Automated verification test bench.

`ewscan/tracker.py`

- `ObservationTracker` : Online memory tracking occupancy, event histories, difference-histogram period deinterleaving, circularly smoothed phase-folded activity profiles, and 13-dimensional per-band feature vectors.

`ewscan/predictor.py`

- `ActivityPredictor` : PyTorch `nn.Module` featuring a 160-unit GRU with residual logit fusion over the tracker prior.
- `OnlinePredictor` : Streaming inference wrapper executing recurrent forward steps in < 0.4 ms.
- `PredictorScheduler` : ML-guided scheduling policy combining GRU forecasts with revisit freshness.
- `train_predictor()` : Offline training pipeline with OneCycleLR and horizon-weighted BCE loss.

`ewscan/rl.py`

- `QNet` : Permutation-equivariant shared per-band encoder with global mean-pooling context head.
- `DQNScheduler` : Real-time scheduling agent combining Double-DQN evaluation with the 32-slot coverage shield.
- `Replay` : Circular experience replay buffer (150,000 transitions).
- `train_dqn()` : Deep Q-network training pipeline featuring model-based demonstration seeding and Polyak target updates.

`ewscan/metrics.py`

- `compute_metrics(env, log)` : Evaluates 13 rigorous figures of merit: P_d , P_{fa} , sensitivity, hit rate, intercept rate, raw & threat-weighted intercept ratio, emitters found, mean intercept time, mean information age, reward, prediction accuracy, and intercept time MAE.

9.2 Service & Web Interface (`server/` and `web/`)

- `server/app.py` : High-performance FastAPI server providing REST endpoints:
 - `GET /api/info` : Receiver specs, preset descriptions, scheduler status.
 - `POST /api/simulate` : Executes full Monte Carlo simulations with selected schedulers; returns truth grids, dwell actions, detections, and metrics.
 - `GET /api/benchmark` : Serves precomputed benchmark results from `reports/benchmark.json`.
 - `GET /api/theory/optimal` : Computes the optimal Vernier revisit period dynamically.
 - `GET /api/training` : Serves training loss and validation convergence logs.
- `web/index.html` & `web/app.js` : Modern interactive single-page application:
 - **Dual Mode UI**: "Simple View" for commanders (plain English status, green/grey catch indicators) and "Detailed View" for EW engineers (full 2D/3D waterfall spectrum, emitter truth overlay, figures of merit tables).
 - **Live A/B Race**: Side-by-side comparative visualization of Receiver A (Conventional Sweep) vs Receiver B (Smart Scan).
 - **Interactive Periodic Intercept Designer**: Real-time visualization of beam coincidences and lock-out zones.

10. Defense & Strategic Impact (SIH / DRDO / BEL Alignment)

1. Atmanirbhar Bharat & Indigenous Innovation:

Delivers a completely indigenous, sovereign AI capability for electronic warfare. Eliminates reliance on pre-mission enemy electronic orders of battle (EOB), allowing Indian armed forces to operate effectively in uncharacterized or contested electromagnetic environments.

2. Zero Hardware Modifications (Pure Software Upgrade):

The Smart Scan architecture requires **no RF hardware redesign**. It operates directly as an algorithmic firmware/software upgrade on existing swept superheterodyne receivers deployed across Indian Air Force, Navy, and Army platforms (e.g., DRDO DLRL / BEL RWR and ESM suites).

3. SWaP-C & Real-Time Feasibility:

The decision cycle requires only **1.3 ms on a standard laptop CPU**—well within the **10 ms dwell budget** of modern fast-tuning local oscillators. No GPU is required at operational inference time.

4. Dual-Use Potential:

Beyond electronic warfare, the underlying cognitive scheduling engine is directly transferable to civilian applications including dynamic spectrum access, cognitive radio networks, anti-jamming communication links, and regulatory spectrum surveillance.

11. Conclusion & Recommendations

The **Smart Scan** system represents a paradigm shift from rigid, pre-planned open-loop sweeps to an intelligent, adaptive, closed-loop Electronic Support receiver. By synthesizing number-theoretic Vernier scheduling, causal Bayesian tracking, residual GRU activity prediction, and safe Deep Reinforcement Learning, the system demonstrates decisive tactical advantages:

- Intercepts **78% more transmission events** and **85% more high-priority threat signals**.
- Eliminates **81% periodic lock-out blind spots**.
- Delivers **15% fresher situational awareness**.

Recommended Roadmap for Field Transition:

1. **Hardware-in-the-Loop (HIL) Integration:** Connect the scheduler via Ethernet/PCIe to an FPGA-controlled RF synthesizer and Software Defined Radio (SDR) test bench.
2. **Extended Domain Randomization:** Incorporate multi-path multipath fading, terrain shadowing, and intentional electronic countermeasures (ECM noise and deceptive jamming).
3. **Multi-Receiver Cooperative Network:** Extend the Q-network context to distributed multi-platform ES nodes, synchronizing distributed dwells to maximize joint probability of intercept.